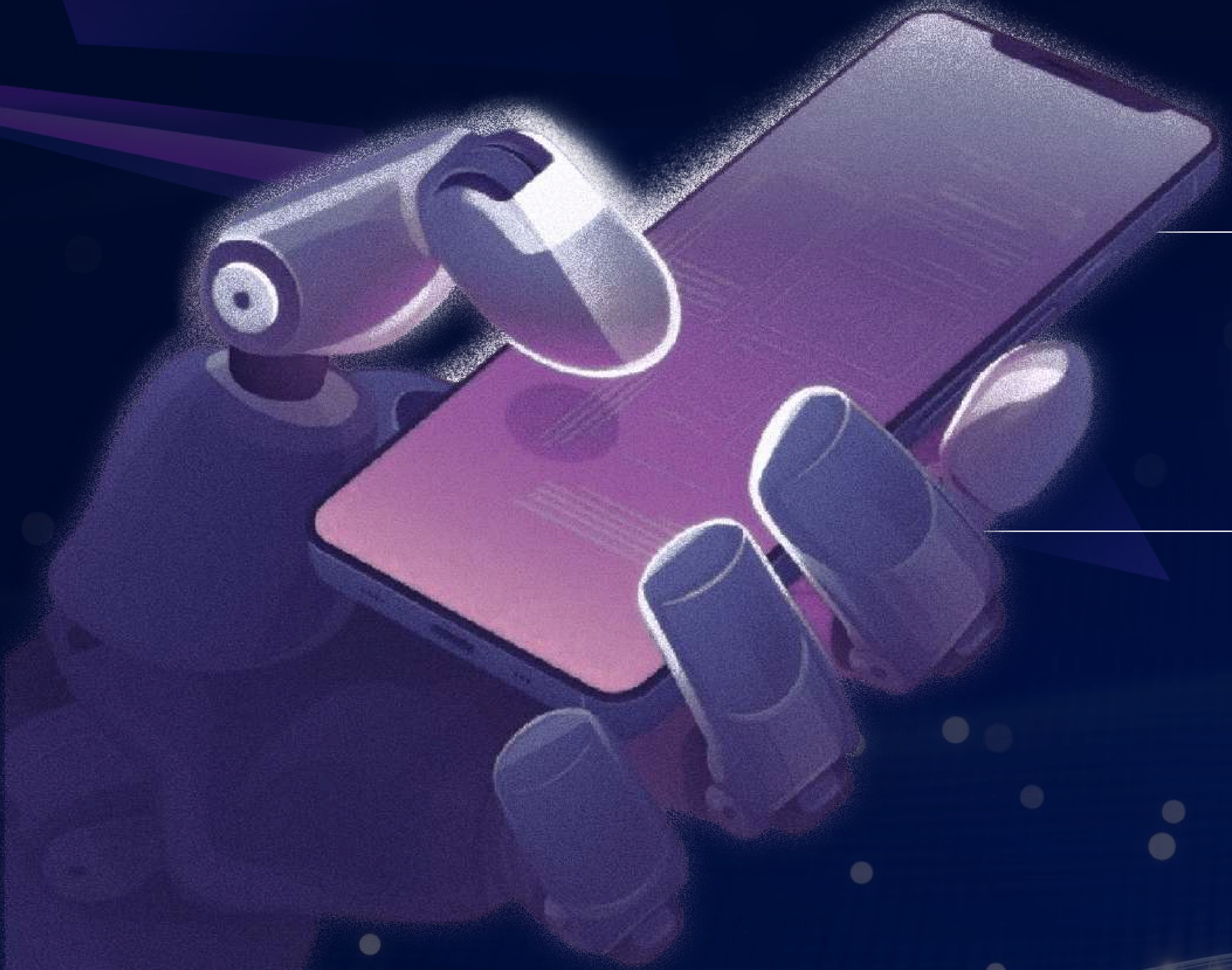

Tips Hindawi Challenge · June–July 2026

NOVELLM

An AI-Powered Reading Companion

Author
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Organization
Edrak for AI

Program
LLMs Internship

Model Core
Mistral-7B · 4-bit

01 / Motivation

READING IS HARD. UNDERSTANDING IS HARDER.

No Instant Q&A

Readers can't ask questions about a book's content without finishing the entire thing first — or searching page by page.

Context Lost Mid-Book

Hard to recall plot points, character arcs, or key events when returning to a book after a break.

Dense Passages

Complex literary or technical language creates friction — especially for non-native readers or students.

Arabic Text Gap

Most AI reading tools only support Latin-script PDFs; Arabic and scanned texts are left completely unsupported.

02 / Solution

WHAT IS NOVELLM?

Upload any PDF book and get instant, AI-grounded insight. Ask questions, generate catch-up summaries, explore character networks, quiz yourself, simplify dense passages — and discover your next read. All in one unified Gradio dashboard, running locally on GPU with a shareable public link.

- Mistral-7B-Instruct-v0.2
- LangChain
- FAISS Vector Store
- Pyvis Graph
- PyMuPDF + EasyOCR+pypdf
- BitsAndBytes 4-bit
- Gradio
- Kaggle GPU

03 / Architecture

HOW IT WORKS

PDF Input
PyMuPDF parser + EasyOCR fallback

->

Chunking
Recursive Character TextSplitter

->

Embeddings
Sentence-transformers → FAISS index

->

Retrieval
Semantic similarity search

->

Generation
Mistral-7B 4-bit inference

->

Dashboard
Gradio UI + ngrok public link

Quantization

NF4 4-bit quantization via BitsAndBytes keeps VRAM usage under ~2GB on Kaggle T4 GPU

Custom Output Parser

Extracts structured JSON (characters, quizzes, recommendations) from raw LLM text output reliably

OCR Fallback

EasyOCR handles Arabic-language and scanned PDFs that PyMuPDF cannot parse digitally

9 CAPABILITIES IN ONE DASHBOARD

PDF Upload & Indexing

Full book ingestion with OCR fallback for Arabic & scanned text

Full Book Q&A

RAG-grounded answers from the actual text, not model memory

Catch-Up Summary

Spoiler-free recap up to your current chapter via slider

Character Network

Auto-extracted character cards + interactive relationship graph

Analytics Dashboard

Word count, reading time, chunk count, mention frequency chart

Excerpt Summarizer

Bulleted summaries of any pasted excerpt or the whole book

Quiz Generator

Auto-generated multiple-choice comprehension questions

Explain This

Simplifies dense or difficult passages into plain language

Book Recommender

Mood-based suggestions by vibe, pacing, ending type, protagonist archetype, length & genre

05 / Deep Dive

THE RAG PIPELINE

01 Text Extraction

PyMuPDF reads the PDF digitally. If that fails (scanned/Arabic), EasyOCR takes over page-by-page.

02 Chunking

RecursiveCharacterTextSplitter splits text into overlapping chunks for semantic coherence.

03 Embedding + Indexing

Sentence-transformers encode chunks; FAISS indexes them for fast similarity search.

04 Retrieval

Query is embedded; top-k most relevant chunks are retrieved from the FAISS index.

05 Generation

Retrieved chunks + user query are passed to Mistral-7B (4-bit) via LangChain pipeline wrapper.

Key Technical Choices

VRAM Usage (4-bit NF4)

~2 GB vs ~14 GB (full precision)

Retrieval Precision

Grounded in source text

Language Support

English + Arabic (OCR)

Why Local RAG?

Answers are grounded in the actual book text — not hallucinated from model weights. Users get page-accurate, spoiler-aware responses.

06 / Deep Dive

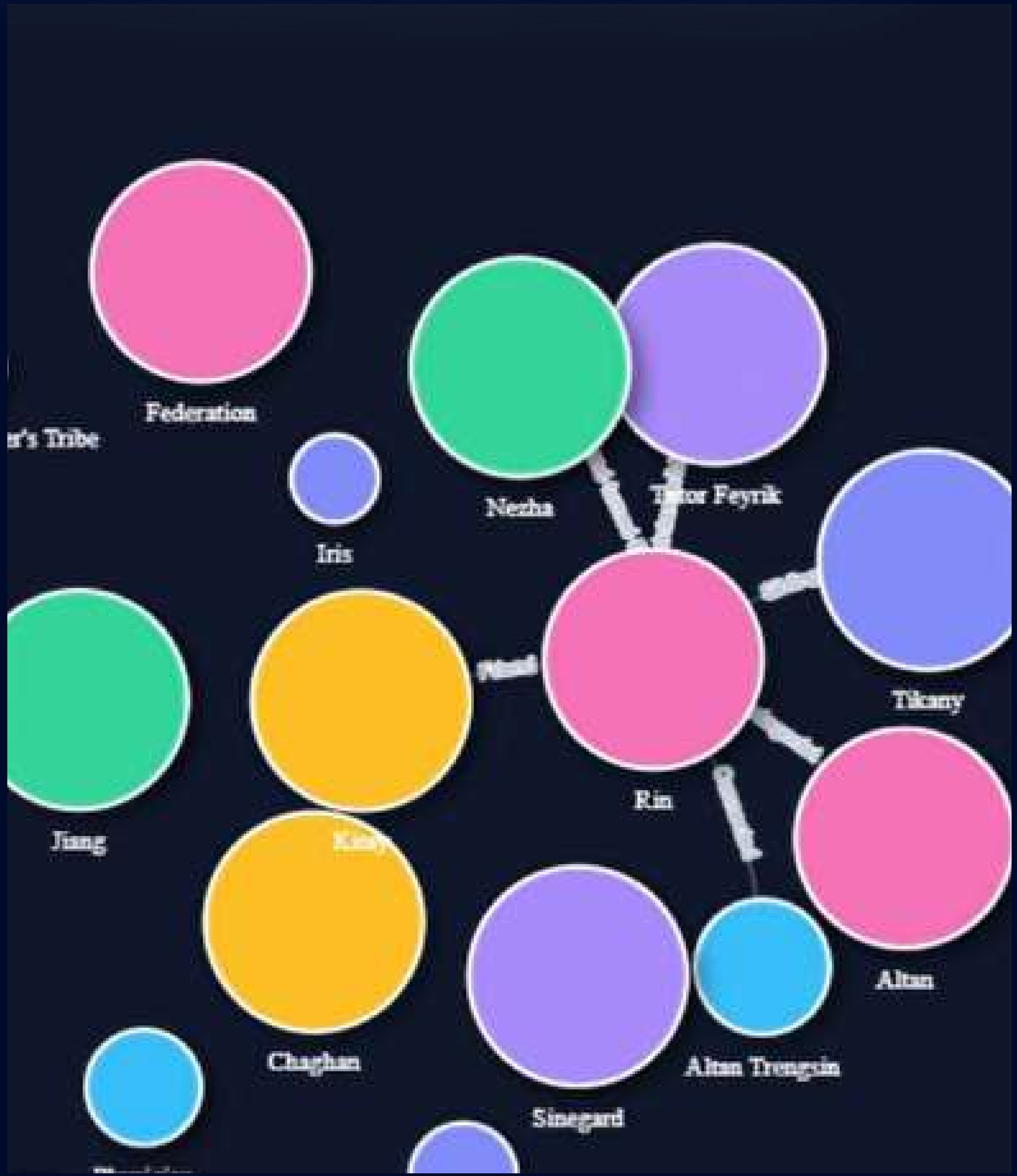
HARACTER NETWORK ANALYSIS

01 LLM Character Extraction
Mistral-7B scans the full text and identifies characters, roles, and relationships

02 Custom Output Parser
Raw LLM text parsed into structured JSON: name, role, mentions, summary, relationships

03 Character Cards
Each character rendered as a clickable card with mention count and role tag

04 Pyvis Relationship Graph
Interactive network graph — nodes = characters, edges = relationships, rendered in browser



07 / Challenge Alignment

LLM CURRICULUM COVERAGE

Mistral-7B Model

Core LLM for all generation tasks — Q&A, summarization, character extraction, quiz generation, and recommendations

RAG (Retrieval-Augmented Generation)

FAISS vector store + LangChain retriever ensures answers are grounded in the actual book text

Summarization

Dedicated excerpt summarizer + spoiler-free catch-up summaries, demonstrating extractive & abstractive summarization

Quantization (4-bit NF4)

BitsAndBytes NF4 quantization applied to Mistral-7B — reduces VRAM from ~14 GB to ~2 GB on Kaggle T4

Output Parser (Chaining)

Custom flexible parser reliably converts raw LLM text → structured JSON for characters, quizzes, and recommendations

Novel Addition: Bilingual + Graph

Arabic OCR support + interactive character relationship graph go beyond curriculum — real-world gaps addressed

WHAT WAS ACHIEVED

<2GB

VRAM under 2 GB

Full Mistral-7B inference running on Kaggle T4 via 4-bit NF4 quantization — no expensive GPU required

9

9 distinct reading features

Q&A · Catch-up · Character network · Analytics · Summarizer · Quiz · Explain This · Recommender — all in one UI

2

Bilingual support (EN + AR)

Automatic EasyOCR fallback handles Arabic & scanned PDFs — a gap most similar tools leave unaddressed

∞

Shareable with zero local setup

Gradio + ngrok tunneling gives any reviewer a live public URL — no installation needed to test

09 / Roadmap

FUTURE IMPROVEMENTS

- Mobile App (iOS / Android)
  Package the reading companion as a native mobile app for on-the-go use — access your book's AI from anywhere
- Domain Fine-Tuning
  Fine-tune Mistral on literary datasets to improve domain knowledge, reduce generic responses, and better understand narrative structure
- **More File Formats**
  Add EPUB, plain text (.txt), and DOCX support beyond the current PDF-only pipeline
- User Accounts & Reading Progress
  Persistent reading history synced across devices — pick up exactly where you left off, across books
- Audio / Text-to-Speech
  Let users listen to summaries, explanations, and catch-ups — turning the companion into an audiobook assistant

THANK YOU

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